ALICE L WALTON SCHOOL OF MEDICINE

Coronary Blood Flow–Vascular Physiology

Last updated February 9, 2026 20 min read ScholarRx

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Adapted from: Coronary Vascular Physiology

Listen to this Brick

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the phasic flow of blood to the ventricular myocardium through an entire cardiac cycle.
2. Discuss the myocardial arterio-venous O_2 difference and oxygen extraction, and the effects of exercise.
3. Explain the regulation of coronary blood flow (autoregulation and metabolic regulation), coronary vascular reserve, coronary steal, and the role of collateral blood vessels in the heart.
4. Explain how autonomic nervous system activation affects coronary vascular resistance and coronary blood flow.

CASE CONNECTION

TR has been having chest pain for the past hour and calls 9-1-1. The paramedics perform a 12-lead ECG, documenting ST depression in the lateral leads. TR's blood pressure is 146/70 mm Hg, and his heart rate is 108/min. The paramedics call the coronary care unit and ask if they should start IV metoprolol, a β-blocker. "Absolutely," is the reply.

You are puzzled and wonder, "Why would we want to start β-blockers?" Consider your answer as you read, and we'll revisit TR at the end of the brick.

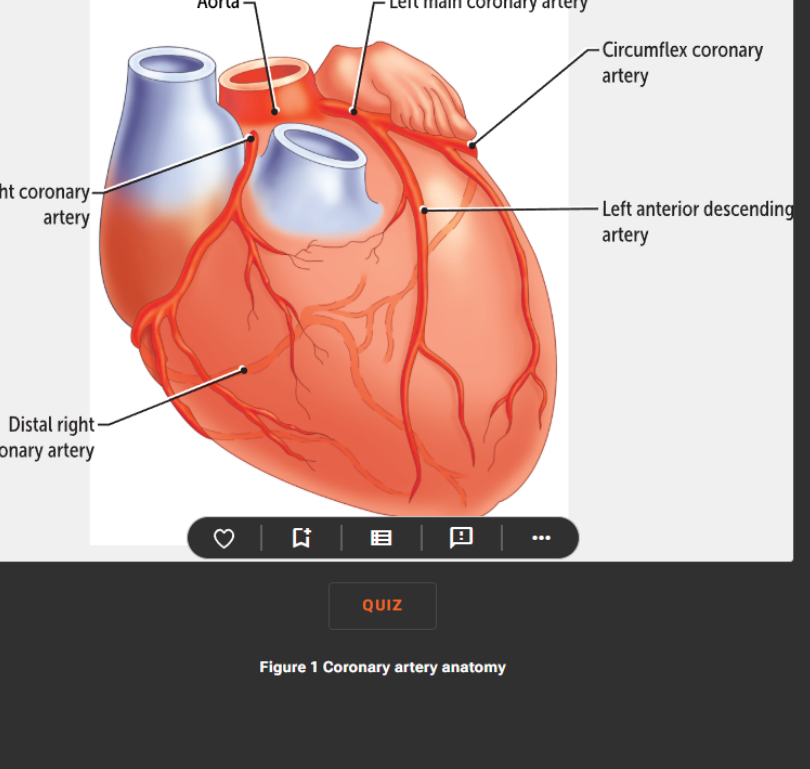
GO TO CONCLUSION ↓

Introduction– blood flow to the heart muscle

Understanding the physiology of blood flow to the heart (coronary vascular) is essential because the heart, despite its size, has a disproportionately high demand for oxygen. It cannot store oxygen and does not obtain oxygen from blood in its chambers; thus, oxygen delivery to the heart is **flow-limited.** Hence, the heart must tightly regulate its blood supply to match metabolic demand or needs in r ecially critical during exercise, stress, tachycardia, or in disease states such as coronary artery disease.

Brief anatomy of coronoary blood flow

To briefly review relevant anatomy, the blood supply to the heart itself is delivered by the right coronary artery (RCA) and left coronary artery (LCA), lying in grooves between the heart's chambers. These epicardial coronary arteries (on the epicardial surface) arise at the root of the aorta, just above the aortic valve (Figure 1).

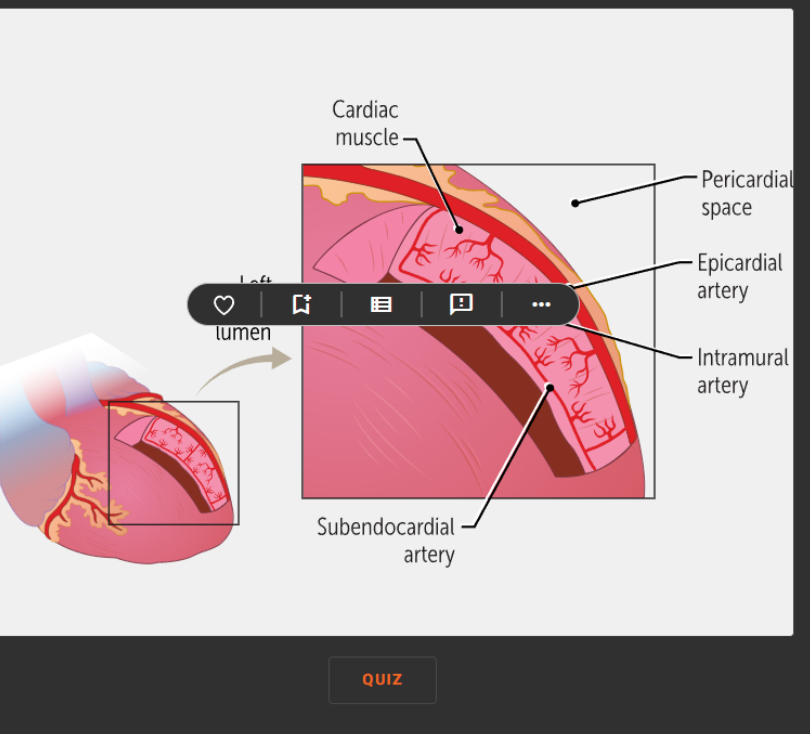

Aorta
Left main coronary artery
Circumflex coronary artery
Right coronary artery
Left anterior descending artery
Distal right coronary artery

QUIZ

Figure 1 Coronary artery anatomy

The arteries run on the outside surface of the heart (epicardium), before branching into smaller branching and penetrating arteries (**intramural arteries**) that serve the mass of the heart, which dive deep into the myocardium. In turn, the intramural arteries give rise to the **subendocardial arteries** (Figure 2). This leads to a well-developed epicardial and subendocardial capillary network.

The capillary density is extremely high in the heart, exceeding 3,000/mm², as compared with 400/mm² in skeletal muscle (Figure 2). This high ratio of capillaries to cardiomyocytes and small diffusion distances results in effective perfusion, with good oxygen delivery, and effective removal of metabolic products (e.g., CO_2 and H^+) at the cardiomyocytes.



QUIZ

Figure 2 Microcirculation in the heart

CLINICAL CORRELATION

Subendocardium and ischemia: The subendocardium (innermost layer of the myocardium) is more susceptible to myocardial ischemia (reduced blood flow) than other heart layers. During systole, the subendocardial arteries experience greater compression from cardiac contraction and ventricular chamber pressure and is the last region to receive blood flow. These flow limitations become more severe in the presence of arterial stenosis, and lower perfusion pressures, further compromising oxygen delivery to this vulnerable region.

The capillaries of the coronary circulation drain deoxygenated blood primarily into the coronary sin m. Some coronary deoxygenated blood flows through thebesian veins (vena cordis minimae) to enter the ventricles directly, allowing deoxygenated blood to circumvent the pulmonary circulation normally required for reoxygenation.

Phasic Supply of Blood to the Heart (LO1)

The perfusion of the heart's capillary network depends on the cardiac cycle phase, particularly whether the ventricles are actively pumping in systole or relaxed in diastole (Figure 3). Myocardial perfusion is reduced during systole and markedly higher during diastole.

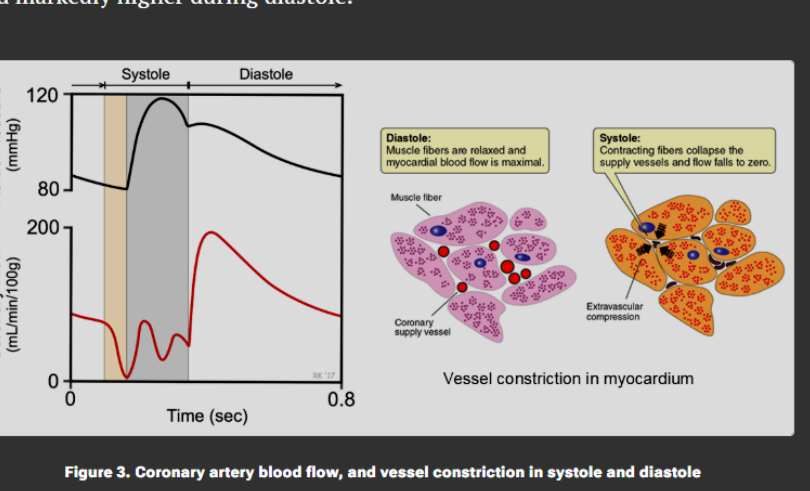


Figure 3. Coronary artery blood flow, and vessel constriction in systole and diastole

During **systole**, ventricular contraction compresses coronary vessels, especially affecting subendocar ntractions are so intense that they greatly increase vascular resistance and partially occlude blood flow through the coronary arteries (Figure 3, right panel). Additionally, open aortic valve leaflets block the coronary artery openings at the aortic root (Figure 4). Thus, the flow of oxygenated blood to supply the myocardium is largely *inhibited during systole.*

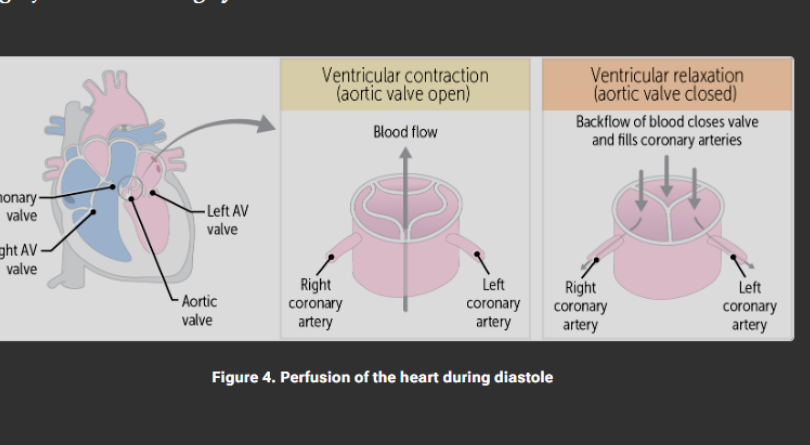


Figure 4. Perfusion of the heart during diastole

What are the first arteries to branch off of the aorta?

In contrast, during **diastole**, relaxation of the myocardium means that the blood vessels in the cardiac muscle themselves are not compressed (Figure 3, see left panel)! You will note that the flow increases at the start of diastole, and then decreases, mirroring the aortic pressure. The majority of coronary blood flow occurs during diastole due to the following reasons:

- **Reduced compressive force** in the relaxed heart results in decreased vascular resistance, maximal coronary perfusion, and increased myocardial blood flow.
- **Opened aortic valves** no longer block the coronary artery opening.

While Figure 3 depicts coronary blood flow in the left ventricle, the pattern is similar for the right ventricle, albeit with lower pressures and thus less pressure differences.

Other Factors affecting Coronary blood flow

Coronary blood flow occurs primarily during diastole, when the heart muscle relaxes, with diastole providing the pressure gradient and duration for filling. The pressure gradient drives blood flow through coronary arteries, while the diastole determines filling time. Anything that alters diastolic pressure or duration will affect coronary perfusion. Left Ventricular End Diastolic Pressure (normally quite low at 5-10 mmHg) acts as an opposing force to blood flow.

Coronary Perfusion Pressure = Diastolic Blood Pressure - Left Ventricular End Diastolic Pressu

The pressure gradient from aorta to coronary arteries drives coronary blood flow and can be compromised by aortic stenosis. In coronary artery disease, the vessels are narrowed, increasing resistance and reducing perfusion pressure and flow. Diseased vessels cannot properly vasodilate in response to vasoactive substances released during metabolic or reactive hyperemia, such as with increased metabolic demands or ischemia. The end result is a reduction in the coronary flow reserve, as discussed later.

CLINICAL CORRELATION

Aortic Stenosis: This valvular obstruction creates a perfect storm for myocardial ischemia through multiple mechanisms. The narrowed aortic valve increases left ventricular workload and oxygen demand while simultaneously reducing coronary perfusion pressure. Additionally, the compensatory left ventricular hypertrophy increases oxygen requirements while compressing intramyocardial vessels. The combination of increased demand and compromised supply explains why angina is a cardinal symptom of severe aortic stenosis, even without significant coronary artery disease.

Heart rate also affects the coronary blood flow. You learned earlier that the heart spends 2/3 of the time in diastole. When the heart rate increases, the resting phase (diastole) shortens - reducing the time when heart muscle receives most of its blood supply - while simultaneously increasing the heart's oxygen demand through more frequent contractions. While healthy hearts partly compensate for tachycardia through metabolic vasodilation (active hyperemia), this is in cur in normal physiology, such as during exercise, but also in disease (as described below).

Fast heart rates (tachycardia) can damage heart muscle, particularly in patients with narrowed coronary arteries who already have limited blood flow to their heart. This double threat of tachycardia and increased oxygen demand is even more dangerous than high blood pressure alone, as hypertension at least improves coronary perfusion pressure despite increasing workload (afterload). In critical situations, such as during heart attacks or surgery, physicians typically administer beta blockers to control heart rate.

CLINICAL CORRELATION

Tachycardia and stenosis: In diseased hearts with stenosis and reduced perfusion pressure, tachycardia can become life-threatening, as the shortened diastole can compromise coronary perfusion, emphasizing the importance of diastolic duration in myocardial oxygen delivery.

When does most of the left coronary blood flow occur?

Supply and demand – O_2 delivery and use (LO2)

Oxygen extraction: Arterio–venous O_2 difference

The heart beats 1 to 3 times per second throughout life, creating a high demand for a constant supply of oxygen. The **arterio-venous (A-V) oxygen difference** represents the amount of oxygen tissues extract from arterial blood before it returns to the heart via veins. The **myocardium oxygen extraction** is extremely high—approximately 70-80% of oxygen is extracted as blood passes through the coronary circulation. For example, the incoming arterial oxygen content (CaO_2 - a product of hemoglobin level and hemoglobin O2 saturation) is 20 mg/dL, while venous oxygen content (CvO_2) is ~ 5 mg/dL, and the resulting in an A-V oxygen difference of 13-15 mg/dL. This represents the highest A-V oxygen difference of any major organ, significantly higher than other organs like skeletal muscle, which extracts only 25-30% of oxygen at rest.

Why is myocardial oxygen extraction so high?

1. **High Oxygen demand:** because the heart is always pumping, and thus has a continuous, high basal oxygen consumption (8-10 ml O_2/min/100g). This demand drives a strong need for oxygen to fuel aerobic metabolism, which is the most efficient way for cardiac cells to generate ATP.
2. **Limited Anaerobic Capacity:** Cardiac muscle requires a substantial amount of oxygen to generate energy through aerobic metabolism.
3. **High Capillary I** capillaries/mm², facilitating efficient oxygen delivery and extraction.

CLINICAL CORRELATION

Flow-Dependent Ischemia: To satisfy its oxidative metabolism energy needs, the heart needs a near uninterrupted O_2 supply. Since the heart is already near maximal O_2 extraction, it relies on increasing blood flow to satisfy demand.

With reduced coronary blood flow to cardiac tissue (e.g., ischemic heart disease, atherosclerosis), myocardial oxygen supply decreases while demand remains unchanged, creating a dangerous imbalance. This mismatch forces cardiac cells to switch from efficient aerobic metabolism to *very limited* anaerobic glycolysis, resulting in the production of less ATP and the generation and accumulation of painful lactate. Thus, even minor coronary occlusions can trigger ischemia, angina, and potentially infarction if flow remains compromised.

Coronary Flow and Myocardial Workload (LO3)

Autoregulation of Coronary Blood Flow

Coronary blood flow is regulated by vessel stretch and the autonomic nervous system, alongside metabolic factors. At rest, coronary flow remains relatively stable despite changes in perfusion pressure, a process called coronary autoregulation. Arterial wall smooth muscle automatically responds to stretch. Increased pressure stretches the blood vessel walls, resulting in vasoconstriction and elps keep blood flow steady, even when mean arterial pressure (MAP) varies between 70-150 mmHg. For reference, normal MAP is around 90 mmHg.

Metabolic control of Coronary Blood flow– hyperemia

The heart muscle extracts a high percentage of oxygen from blood already, so increasing metabolic demand is only met by increasing blood flow (Figure 5). Coronary blood flow closely parallels the heart's oxygen needs and workload. When cardiac activity increases, coronary blood flow increases proportionately through active/reactive/metabolic hyperemia. This occurs because the myocardium releases vasodilatory substances when oxygen demand rises or supply falls, causing coronary arteries to relax and dilate, reducing arterial resistance and increasing coronary blood flow to fulfill the increased demand for oxygen.

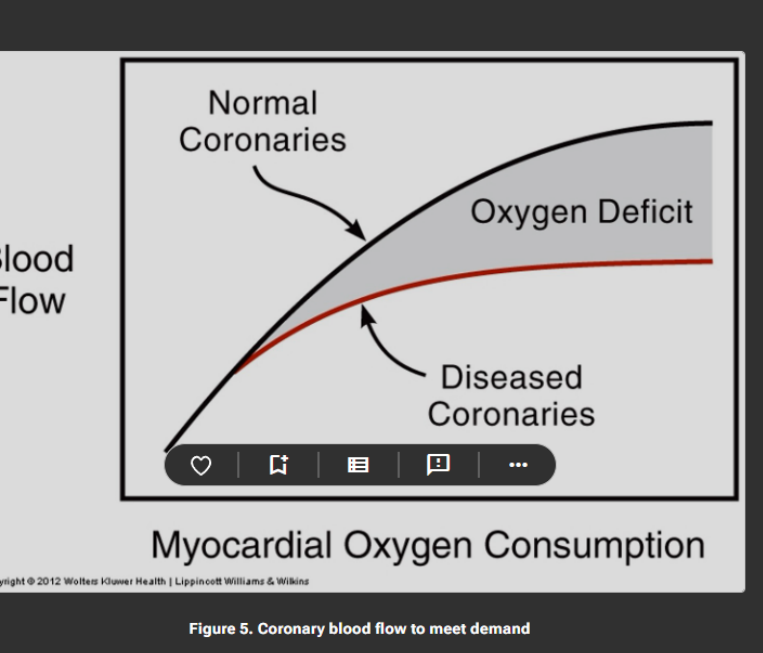


Figure 5. Coronary blood flow to meet demand

Remember that blood flow equals pressure change divided by resistance (Q = ΔP/R), so reduced resistance through vasodilation significantly increases coronary blood flow. Three key factors drive this coronary dilation:

- **Increase in adenosine.** As ATP is consumed, adenosine monophosphate (AMP) forms; adenosine is a byproduct of ATP breakdown and is increased during states of high energy consumption. Adenosine is a strong vasodilator.
- **Decrease in adenosine triphosphate (ATP).** This results from high cardiac energy consumption. When ATP levels are low, ATP-sensitive potassium (K_{ATP}) channels open, causing the vascular smooth muscle to hyperpolarize.
- **Increase in nitric oxide (NO).** Produced by endothelial cells in response to shear stress, facilitating vessel relaxation. As cardiac output increases, blood flows faster through the arteries, resulting in higher shear forces on the endothelial cells, which causes them to respond by secreting nitric oxide (NO).

Other factors that may contribute to coronary vasodilation alongside these "big three" include reduced tissue oxygen and increased potassium, hydrogen ions, carbon dioxide, and vasodilatory prostaglandins—all typical products of active muscle metabolism. nation, inhibits force generation by the contractile proteins, and affects calcium fluxes.

Coronary Flow Reserve

Coronary flow reserve is the difference between maximum flow during vasodilation and resting flow. It is essentially a safety factor. Healthy individuals have sufficient reserve to meet increased demands, whereas atherosclerosis reduces this reserve, potentially causing ischemia during stress when diseased arterioles cannot dilate further to increase flow, and supply does not meet demand (Figure 5).

CLINICAL CORRELATION

The principle of flow-dependent perfusion of the myocardium is the basis for **cardiac stress testing.** In a controlled, monitored environment, the cardiac workload is increased by exercise (treadmill or stationary bike) or use of dobutamine, a β_2 agonist with positive inotropic and chronotropic effects (increased contractility and heart rate). This assesses the coronary flow reserve.

When cardiac workload increases, regions of the myocardium perfused by atherosclerotic vessels may have minimal ability to dilate and tap into the coronary flow reserve. Critical coronary artery stenosis can become unmasked, with findings of chest pain, EKG changes, regional wall motion abnormalities on echocardiography, or reversible perfusion defects on nuclear medicine imaging.

A negative dobutamine stress echo in a patient who achieves the target heart rate correlates with a 1.3% risk of an acute cardiac event (myocardial infarction or cardiac death) in the next 5 years. A positive DSE typically warrants coronary angiography to visu presence and severity of blockages. A decision is then made on the best course of action to decrease the risk of acute cardiac events: Percutaneous coronary intervention (stenting), surgery (coronary artery bypass grafting), or best medical therapy.

see references: Poldermans

Why is diastole shorter during exercise?

Collateral blood vessels

The **absence of robust intracoronary or collateral vascular connections** between the main arterial supplies means that blockage of an artery leads to ischemia in its territory. **Abrupt** occlusion of a coronary artery or its branches leads to reduced blood flow (ischemia) in that region and produces necrosis (myocardial infarctio

However, if the narrowing is **slow and gradual, collateral vessels** can form (angiogenesis) and partially compensate for reduced oxygen and nutrient delivery to the affected area. Angiogenesis and collateralization within the myocardium can develop, especially with chronic ischemic coronary artery disease, creating parallel vessels that allow blood to partially bypass blockages.

Coronary Steal – stolen flow from ischemic regions

Coronary artery steal occurs in the context of coronary artery disease with narrowed large epicardial arteries and areas of ischemic myocardium. This typically occurs with the administration of *arteriole-dilating* drugs used for cardiac stress testing (e.g. adenosine). Arterioles in the ischemic region are already maximally dilated, but those in the non-ischemic areas dilate enough to divert blood flow away from the ischemic site. Nitrates are anti-anginal medications. They cause venodilation (decreasing myocardial oxygen demand) and *dilation of larger epicardial arteries* and collaterals. Blood flow is globally increased, and the region of ischemia maintains its greater arteriolar dilation relative to the healthy myocardium, preventing steal.

What triggers coronary vasodilation during increased myocardial activity?

CLINICAL CORRELATION

Angina Pectoris: literally means chest strangling. This classic chest pain occurs when myocardial oxygen demand exceeds restricted supply, and the cardiac muscle becomes ischemic. Angina results from ischemia-induced release of algogenic substances (substances that give a pain signal). The characteristic radiation to the left arm occurs because cardiac chemosensitive afferents converge with somatic nerve pathways in the spinal cord.

There are different types of angina: unstable angina, which occurs with exertion and is predictable (mild coronary occlusion), and stable angina, which occurs at rest (demand > supply, severe coronary occlusion). The third form, vasospastic angina, results from coronary vasospasm and is not resolved by rest (e.g., watching TV).

How does the m ary blood flow?

Autonomic nervous system and the heart (LO4)

Sympathetic nervous system (SNS): The nervous system does not have a major role in coronary blood flow. Recall that the increased local flow by metabolites (active hyperemia) is due to increased heart activity via β1 adrenergic receptors. The overriding of the sympathetic nervous system by local control of blood flow is termed "**functional sympatholysis**". You would not want to have decreased blood flow to the heart during "fight or flight" such as when running away from danger!

The activation of the SNS causes the coronary arteries to constrict via the α_1-**adrenergic** receptors, and this restriction should reduce blood flow. However, the α_1 effect on coronary arteries is far less potent than its effect on peripheral arteries. Coronary arteries have a minimal number of α_1 receptors, and these are easily overridden by the vasodilator metabolites as described above. The smaller coronary vessels responsible for resistance have heavy β_2 innervation and respond well to circulating epinephrine, resulting in vasodilation due to smooth muscle relaxation (the same is true in skeletal muscle resistance arteries). The increased cardiac activity due to the β1 effects of catecholamines will also *secondarily* increase coronary vasodilation due to increased levels of adenosine and decre

Parasympathetic Nervous System (PSNS): The PSNS stimulates the muscarinic cholinergic receptors on the coronary vessels, although this effect is minor and leads to only slight vasodilation. Note that the majority of the PSNS innervation is of the SA node. However, if the PSNS causes a significantly large decrease in heart rate, the decreased metabolic demand and lowered metabolic vasodilators would result in vasoconstriction.

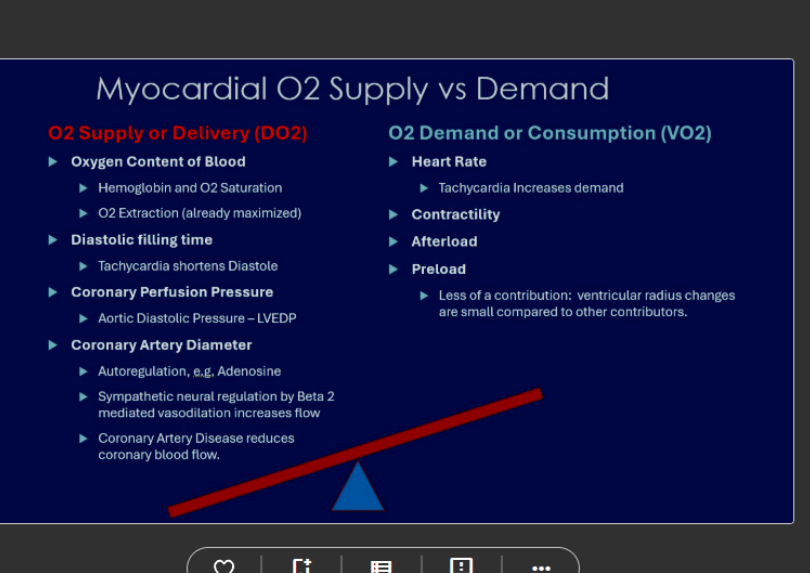
ACTIVE LEARNING QUESTION

Choose four factors that increase Myocardial Oxygen Supply, leaving four factors that increase Myocardial Oxygen Demand.

A. ☑ Oxygen content of blood (Hemoglobin level and SaO2)
B. ☐ Tachycardia
C. ☐ Myocardial wall tension
D. ☑ Coronary perfusion pressure
E. ☑ Diastolic time
F. ☐ Contractility
G. ☐ Afterload
H. ☑ Adenosine

Explanation

Myocardial Oxygen Supply is positively correlated with oxygen content of blood (A), coronary perfusion pressure (especially higher diastolic pressure) (D), time spent in diastole (E), and vasodilation (H). Demand is positively correlated with increasing heart rate (due to both increased workload and decreased time in diastole) (B), wall tension (increased preload resulting in increased Frank-Starling effects) (C), contractility (F), and afterload (G). **(Figure 6)**



Myocardial O2 Supply vs Demand

O2 Supply or Delivery (DO2)
- Oxygen Content of Blood
  - Hemoglobin and O2 Saturation
  - O2 Extraction (already maximized)
- Diastolic filling time
  - Tachycardia shortens Diastole
- Coronary Perfusion Pressure
  - Aortic Diastolic Pressure – LVEDP
- Coronary Artery Diameter
  - Autoregulation, e.g. Adenosine
  - Sympathetic neural regulation by Beta 2 mediated vasodilation increases flow
  - Coronary Artery Disease reduces coronary blood flow.

O2 Demand or Consumption (VO2)
- Heart Rate
  - Tachycardia increases demand
- Contractility
- Afterload
- Preload
  - Less of a contribution: ventricular radius changes are small compared to other contributors.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to TR, why would β-blockers be beneficial in his management?

You consider that, in the setting of a myocardial infarction, it is critical to both decrease the myocardial workload and maintain optimal blood flow through the coronary arteries. β-Blockers do both. First, β-blockers lower heart rate, resulting in more time for diastolic filling. Because most of the blood flow to the coronary arteries occurs in diastole, blood flow to the myocardium will increase.

Second, β-blockers decrease myocardial oxygen demand by decreasing heart rate, blood pressure (systemic vascular resistance or afterload), and contractility of the heart.

Summary

- The heart has a high oxygen demand and relies on coronary blood flow to meet its metabolic needs (demand vs supply).
- Coronary blood flow is primarily regulated during diastole, with reduced flow during systole due to ventricular contraction compressing coronary vessels.
- The open aortic valve blocks blood flow to the coronary vessels.
- With tachycardia, the heart has an increased workload and spends less time in diastole, so there is less time available for coronary perfusion. This increases the risk of ischemia, especially in diseased hearts with stenosis.
- The myocardium ilitates efficient oxygen delivery and extraction with a high arterio-venous oxygen difference.
- Coronary blood flow is controlled by metabolic factors released or consumed during cardiac muscle metabolism. This includes adenosine, nitric oxide, and low ATP, which trigger vasodilation during increased myocardial activity.
- Decreases in ATP and increases in adenosine and nitric oxide are the primary factors that trigger vasodilation. This allows for increases in oxygen delivery to the heart during periods of increased activity.
- Autoregulation causes coronary vessels to dilate in response to decreases in blood flow so that blood flow to the heart is maintained during resting periods.
- The myogenic reflex causes the vasoconstriction of coronary arteries when those arteries are stretched because of an increase in pressure.
- Coronary flow reserve is the difference between maximum and resting flow. Atherosclerosis reduces this reserve and potentially causes ischemia during stress.
- Sympathetic α1-adrenergic receptors cause vasoconstriction of the coronary vessels. This effect is easily overridden by metabolic vasodilation.
- The heart's high oxygen demand necessitates precise regulation of coronary blood flow, which is primarily flow-limited and must match metabolic needs in real-time.

Review Questions

1. During which phase of the cardiac cycle does the area represented in this infarction receive i ary arteries?

   A. ○ Atrial diastole
   B. ○ Ventricular contraction
   C. ● Ventricular diastole
   D. ○ Ventricular systole

Hide Explanation

The correct answer is (C). The patient has a left ventricular anterior wall MI, represented by ST elevation in leads V_3 and V_4. During ventricular diastole, the ventricles are relaxed and not compress the coronary vessels. This allows for maximal blood flow to the heart. Incorrect answers: Atrial diastole (A) occurs during ventricular systole, which is when the ventricles are contracting. So, flow to the coronary arteries, arterioles, and capillaries is reduced. Ventricular contraction (B) compresses the coronary vessels, reducing flow; flow is especially reduced to the left coronary artery, because the left ventricle is contracting so intensely. Maximal coronary blood flow occurs during ventricular diastole, and it varies during the cardiac cycle. During ventricular systole (D), ventricular contraction compresses the coronary vessels, reducing flow.

2. Which of the following actions is mostly likely to cause coronary artery constriction?

   A. ○ Adenosin
   B. ● α_1-Adrenergic receptor stimulation
   C. ○ β_2-Adrenergic receptor stimulation
   D. ○ Breakdown of ATP
   E. ○ Nitric oxide release

Hide Explanation

The correct answer is (B). Stimulation of these receptors by the sympathetic nervous system results in vasoconstriction in all arteries, increasing the resistance in the vessels. Note, this can be overridden by vasodilator metabolites released by the heart to cause active hyperemia. Incorrect answers: Adenosine release (A) causes vasodilation. Stimulation of β_2-adrenergic receptors (C) causes vasodilation. Breakdown of ATP (D) causes adenosine production, which results in vasodilation. Nitric oxide release (E) allows nitric oxide to diffuse into smooth muscle cells, causing vasodilation.

3. A 55-year-old man is undergoing treadmill exercise testing. During exertion, myocardial oxygen demand increases substantially. What changes occur in the cardiac system to meet increased demand?

   A. ○ Decreased arteriovenous oxygen difference
   B. ○ Increased hemoglobin affinity for oxygen
   C. ○ Increased oxygen extraction by cardiac muscle
   D. ○ Coronary vasoconstriction
   E. ● Coronary vasodilation

Hide Explanation

The correct answer is (E). Coronary vasodilation increases flow, and since oxygen extraction (A-V O_2 difference) is already maximal in cardiac tissues, this is the only effective way to meet increased myocardial oxygen demand. Incorrect answers: (A) The arteriovenous O_2 difference is already near maximal at rest and does not decrease. (B) Hemoglobin affinity does not increase; in fact, exercise and the increased CO2 production (lowered pH) shift the curve right (Bohr effect) to release more O_2 in exercising tissues. (C) Extraction is already ~70–80% at baseline, leaving little reserve to increase extraction. (D) Coronary vasoconstriction would reduce flow and worsen ischemia.

4. A 65-year-old man with a history of intermittent chest pain undergoes a pharmacologic adenosine nuclear stress test. He develops chest pain during the adenosine injection and the nuclear imaging study reveals poor perfusion of a region of myocardium. The chest pain resolves and the perfusion defect resolves a few minutes later. Which of the following explains the temporary ischemia?

   A. ○ Decreased
   B. ● Enhanced coronary steal phenomenon
   C. ○ Smooth muscle dilation due to stretch
   D. ○ Increased α_1-adrenergic–mediated vasoconstriction

Hide Explanation

Correct answer is (C). In coronary artery disease, narrowed vessels reduce blood flow and cause ischemic areas of the heart. These ischemic affected regions release vasodilators, so their arterioles are already maximally dilated at rest. With adenosine infusion, the arterioles in the healthy regions of heart dilate, pulling blood flow away from the ischemic region. Incorrect answers: (A) Oxygen extraction is already near-maximal in the myocardium at rest. The heart can only meet oxygen demands by increasing coronary blood flow. (C) Autoregulation adjusts arteriolar vessel diameter by relaxing smooth muscles when stretched to create a stable flow. It does not cause ischemia. (D) α_1-mediated vasoconstriction does not significantly occur in the coronary circulation because local metabolic vasodilators override the sympathetic response.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the phasic flow of blood to the ventricular myocardium through an entire cardiac cycle.~~

02 ☑ ~~Discuss the myocardial arterio-venous O_2 difference and oxygen extraction, and the effects of exercise.~~

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Objective Confidence: 4/4

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